

Traveling Salesperson Problem (TSP) Reinforcement Learning Implementation

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1 Introduction

This project implements a Reinforcement Learning agent using the **Proximal Policy Optimization (PPO)** algorithm to solve the Traveling Salesperson Problem on a 10×10 adjacency matrix. The environment is built using the OpenAI Gymnasium framework.

2 Setup and Installation

2.1 Environment Configuration

To isolate dependencies, it is recommended to use a virtual environment:

```
1 # Create the virtual environment
2 python -m venv .venv
3
4 # Activate on Windows
5 .venv\Scripts\activate
6
7 # Update pip
8 python -m pip install --upgrade pip
```

2.2 Dependency Installation

Install the core libraries required for the Reinforcement Learning loop and visualization:

```
1 pip install stable-baselines3 gymnasium pyyaml networkx
   matplotlib numpy tensorboard
```

3 Execution Workflow

3.1 Training

The `train.py` script initializes the environment and trains the PPO agent. Training logs are stored in the `/logs` directory and can be visualized via TensorBoard.

```
1 python train.py
```

3.2 Evaluation

To test the trained model and visualize the pathing logic on the graph, execute:

```
1 python test.py
```

4 Key Technical Logic

4.1 Normalization

For numerical stability in the neural network, the adjacency matrix M is normalized by a factor of 10. The observed cost c' is defined as:

$$c' = \frac{c}{10}$$

where c is the raw integer cost (1–10). A cost of 0.0 represents an inaccessible node.

4.2 Reward Function

The agent receives a reward R based on the following logic:

- **Invalid Move:** -20.0 and episode termination.
- **Goal City:** $20.0 - c'$.
- **Non-Goal Move:** $-c'$.
- **Completion:** $+50.0$ bonus when all required cities are visited.

5 Project Directory Structure

- `tsp_env.py`: The custom Gymnasium environment.
- `data_loader.py`: Handles configuration and matrix generation.
- `config.yaml`: Hyperparameters and environment seeds.
- `output/matrix_log.txt`: Reference log for the current matrix.